

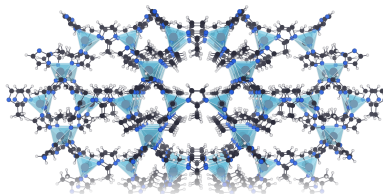


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Molecular Dynamics with CRYSTAL

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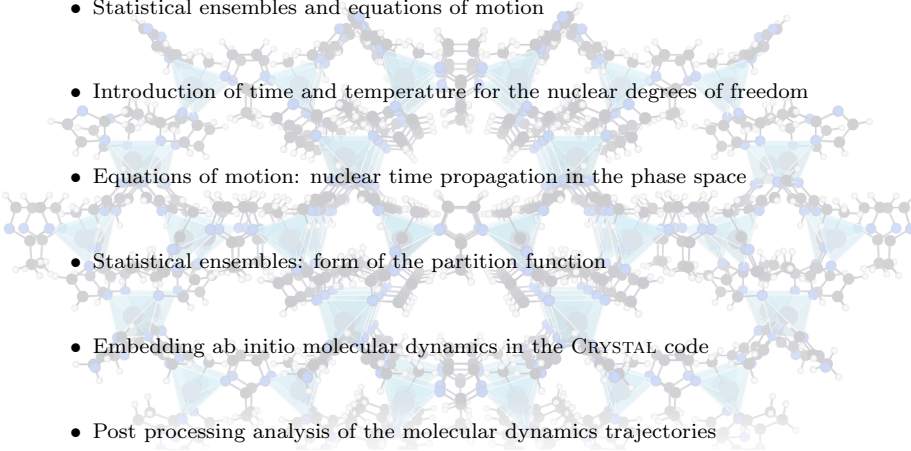
Theoretical Chemistry Group
University of Torino



MSSC2025
TORINO 8-12 SEP 2025

Outline of the Talk

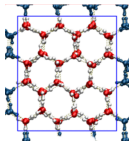
Ab initio Molecular Dynamics in the CRYSTAL code

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- Statistical ensembles and equations of motion
 - Introduction of time and temperature for the nuclear degrees of freedom
 - Equations of motion: nuclear time propagation in the phase space
 - Statistical ensembles: form of the partition function
 - Embedding ab initio molecular dynamics in the CRYSTAL code
 - Post processing analysis of the molecular dynamics trajectories

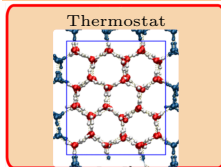
Statistical Ensembles and Equations of Motion

A **statistical ensemble is a collection of microstates** that are compatible with a specified macrostate of a thermodynamic system. Each ensemble is described by a set of independent variables related to the macroscopic state that are kept constant during time evolution of the system, while all the other quantities can fluctuate.

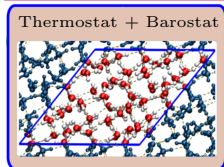
Microcanonical
NVE



Canonical
NVT



Isothermal-Isobaric
NPT



where N is the number of atoms, V is the volume of the system, E the internal energy, T is the temperature of the reservoir and P the external pressure. Given a point Γ in the **phase space**, a statistical ensemble is described by a specific **partition function**

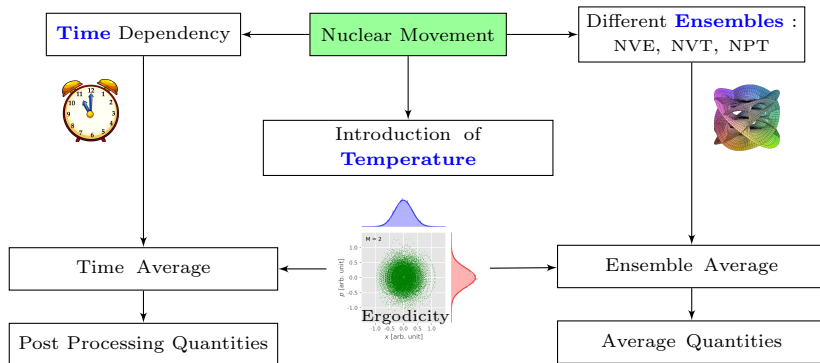
$$Z = \zeta \int d\Gamma e^{-w(\Gamma)} f(\Gamma) \quad \text{with} \quad w(\Gamma) = \int \nabla_{\Gamma} \cdot \dot{\Gamma}(t) dt$$

The exponential term is related to the **equations of motion** $\dot{\Gamma}(t) = \frac{\partial \Gamma}{\partial t}$

The form $f(\Gamma)$ is the **distribution function** related to the probabilities that the system is found in a given allowed microstate and to the equations of motion conserved quantities

Molecular Dynamics : Introduction of Time and Temperature

↗ The molecular dynamics method is based on the introduction of a **dynamic formulation for the nuclear degrees of freedom** $\mathbf{q}, \mathbf{p} = \{\mathbf{q}_\mu, \mathbf{p}_\mu\}$ ($\mu = 1, \dots, N$)



↗ Definition of a set of **classical equations of motion for the nuclei**

↗ Calculation of **average quantities over the ensemble** sampled by the system



The **time evolution** of a function $A[\Gamma(t)]$ that only depends on the generalized phase space coordinates $\Gamma \equiv \Gamma(t) = [\kappa_1(t), \dots, \kappa_n(t)]$ and not explicitly on time, is given by

$$\frac{d}{dt} A[\Gamma(t)] = \sum_{\alpha=1}^n \frac{\partial A[\Gamma(t)]}{\partial \kappa_{\alpha}(t)} \cdot \dot{\kappa}_{\alpha}(t) = \frac{\partial A[\Gamma(t)]}{\partial \Gamma(t)} \cdot \dot{\Gamma}(t) \equiv i\mathcal{L} A[\Gamma(t)]$$

where the **Liouville operator** $\mathcal{L}$ has been defined as

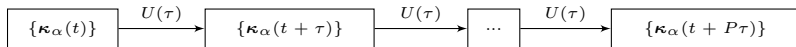
$$i\mathcal{L} = \dot{\Gamma}(t) \cdot \frac{\partial}{\partial \Gamma(t)} = \frac{d\Gamma(t)}{dt} \cdot \frac{\partial}{\partial \Gamma(t)}$$

$$\frac{d\Gamma(t)}{dt} = \left(\frac{d\kappa_1}{dt}, \dots, \frac{d\kappa_n}{dt} \right) \rightarrow \text{equations of motion}$$

Applying this formulation to describe the time evolution of a point in phase space leads to

$$\frac{d}{dt} \Gamma(t) = i\mathcal{L} \Gamma(t) \quad \rightarrow \quad \Gamma(t + \tau) = e^{i\mathcal{L} \tau} \Gamma(t) \approx U(\tau) \Gamma(t)$$

$$U(\tau) = e^{i\mathcal{L} \tau} \approx e^{i\mathcal{L}_n \tau/2} \dots e^{i\mathcal{L}_2 \tau/2} e^{i\mathcal{L}_1 \tau} e^{i\mathcal{L}_2 \tau/2} \dots e^{i\mathcal{L}_n \tau/2} + \mathcal{O}(\tau^{2n-1})$$





Example. Starting from a simple Hamiltonian with the form (with $\mathbf{q} = \{\mathbf{q}_\mu\}$ and $\mathbf{p} = \{\mathbf{p}_\mu\}$ where $\mu = 1, \dots, N$ and $\mathbf{p}_\mu, \mathbf{q}_\mu$ are three dimensional vectors)

$$\mathcal{H}(\mathbf{p}, \mathbf{q}) = \sum_{\mu=1}^N \frac{\mathbf{p}_\mu^2}{2m_\mu} + E_{e,0}(\mathbf{q}) \quad \longrightarrow \quad \begin{cases} \dot{\mathbf{q}}_\mu = \frac{\partial \mathcal{H}}{\partial \mathbf{p}_\mu} = \frac{\mathbf{p}_\mu}{m_\mu} \\ \dot{\mathbf{p}}_\mu = -\frac{\partial \mathcal{H}}{\partial \mathbf{q}_\mu} = \mathbf{F}_\mu \end{cases} \quad \mu = 1, \dots, N$$

$$\Gamma(t) = [\mathbf{q}(t), \mathbf{p}(t)] \quad \rightarrow \quad \frac{d\Gamma(t)}{dt} = \sum_{\mu=1}^N \left(\frac{d\mathbf{q}_\mu}{dt} \cdot \frac{\partial}{\partial \mathbf{q}_\mu} + \frac{d\mathbf{p}_\mu}{dt} \cdot \frac{\partial}{\partial \mathbf{p}_\mu} \right) \Gamma(t) \equiv i\mathcal{L} \Gamma(t)$$

The Liouville operator can be written as the sum of two contributions

$$i\mathcal{L} = i\mathcal{L}_q + i\mathcal{L}_p \quad \text{where} \quad i\mathcal{L}_q = \sum_{\mu=1}^N \frac{\mathbf{p}_\mu}{m_\mu} \cdot \frac{\partial}{\partial \mathbf{q}_\mu} \quad \text{and} \quad i\mathcal{L}_p = \sum_{\mu=1}^N \mathbf{F}_\mu \cdot \frac{\partial}{\partial \mathbf{p}_\mu}$$

Two different time propagators can then be defined as

$$\Gamma(t + \tau) = [\mathbf{q}(t + \tau), \mathbf{p}(t + \tau)] \approx \left[e^{i\mathcal{L}_p \tau/2} e^{i\mathcal{L}_q \tau} e^{i\mathcal{L}_p \tau/2} \right] [\mathbf{q}(t), \mathbf{p}(t)] + \mathcal{O}(\tau^3)$$

$$\Gamma(t + \tau) = [\mathbf{q}(t + \tau), \mathbf{p}(t + \tau)] \approx \left[e^{i\mathcal{L}_q \tau/2} e^{i\mathcal{L}_p \tau} e^{i\mathcal{L}_q \tau/2} \right] [\mathbf{q}(t), \mathbf{p}(t)] + \mathcal{O}(\tau^3)$$



$$i\mathcal{L} = i\mathcal{L}_q + i\mathcal{L}_p \quad \text{where} \quad i\mathcal{L}_q = \sum_{\mu=1}^N \frac{\mathbf{p}_\mu}{m_\mu} \cdot \frac{\partial}{\partial \mathbf{q}_\mu} \quad \text{and} \quad i\mathcal{L}_p = \sum_{\mu=1}^N \mathbf{F}_\mu \cdot \frac{\partial}{\partial \mathbf{p}_\mu}$$

Velocity Verlet updating scheme for time propagation :

$$\mathbf{\Gamma}(t + \tau) = [\mathbf{q}(t + \tau), \mathbf{p}(t + \tau)] \approx e^{i\mathcal{L}_p\tau/2} e^{i\mathcal{L}_q\tau} e^{i\mathcal{L}_p\tau/2} [\mathbf{q}(t), \mathbf{p}(t)] + \mathcal{O}(\tau^3)$$

Directly coded in the CRYSTAL code ... ($\mu = 1, \dots, N$)

Starting point (initial conditions) : $\mathbf{q}_\mu(t), \mathbf{p}_\mu(t), \mathbf{F}_\mu[\{\mathbf{q}_\mu(t)\}]$

Step 1. Propagator $e^{i\mathcal{L}_p\tau/2}$: $\mathbf{p}_\mu(t + \tau/2) \leftarrow \mathbf{p}_\mu(t) + \frac{\tau}{2} \mathbf{F}_\mu[\{\mathbf{q}_\mu(t)\}]$

Step 2. Propagator $e^{i\mathcal{L}_q\tau}$: $\mathbf{q}_\mu(t + \tau) \leftarrow \mathbf{q}_\mu(t) + \frac{\tau}{m_\mu} \mathbf{p}_\mu(t + \tau/2)$

Step 3. Updating forces : compute $\mathbf{F}_\mu[\{\mathbf{q}_\mu(t + \tau)\}]$

Step 4. Propagator $e^{i\mathcal{L}_p\tau/2}$: $\mathbf{p}_\mu(t + \tau) \leftarrow \mathbf{p}_\mu(t + \tau/2) + \frac{\tau}{2} \mathbf{F}_\mu[\{\mathbf{q}_\mu(t + \tau)\}]$

Molecular Dynamics : Generalized description of the ensembles

Ensemble →	NVE	NVT	NPT
	Sys.	Sys. + Th.	Sys. + Th. + Bar.
Coordinates	$\mathbf{p}, \mathbf{q}$	$\mathbf{p}, \mathbf{q}, p_\eta, \eta$	$\mathbf{p}, \mathbf{q}, p_\eta, \eta, p_v, V$
Hamiltonian	$\mathcal{H}(\mathbf{p}, \mathbf{q})$	$\mathcal{H}(\mathbf{p}, \mathbf{q}, p_\eta, \eta)$	$\mathcal{H}(\mathbf{p}, \mathbf{q}, p_\eta, \eta, p_v, V)$
Liouville Operators	2	5	10
Conserved quantity (3D)	$N, V, E, \mathbf{P}$	$N, V, T, e^\eta \mathbf{P}$	$N, P, T, V^{1/d} e^\eta \mathbf{P}$
Partition function (3D)	$Z(N, V, E, \mathbf{0})$	$Z(N, V, T, \mathbf{0})$	$Z(N, P, T, \mathbf{0})$

N = number of atoms

V = volume

E = total energy

T = **temperature**

P = **pressure**

$\mathbf{P}$ = total linear momentum

Sys. = system

Th. = **thermostat**

Bar. = **barostat**

$\mathbf{q}, \mathbf{p}$ = nuclear positions and momenta

η, p_η = **thermostat degrees of freedom**

V, p_v = **barostat degrees of freedom**



Consider a general case in which there are n_c conservation laws of the form

$$\Lambda_k(\mathbf{\Gamma}) = C_k \quad \text{which satisfy the condition} \quad \frac{d\Lambda_k}{dt} = 0 \quad k = 1, \dots, n_c$$

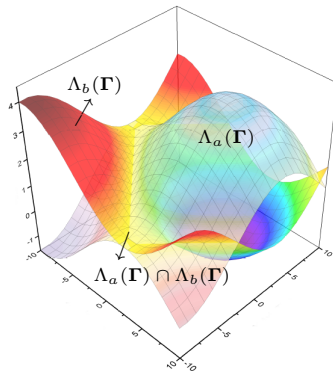
where $\{C_k\}$ defines a set of constants. The distribution generated by the dynamics samples the intersection of the hypersurfaces (subspace of the whole phase space) represented by all of the conservation laws, that is

Distribution function:
$$f(\mathbf{\Gamma}) = \prod_{k=1}^{n_c} \delta(\Lambda_k(\mathbf{\Gamma}) - C_k)$$

General procedure

to define the form of the distribution function:

- ↪ determine the degrees of freedom that define the point in phase space $\mathbf{\Gamma}$
- ↪ form of the Hamiltonian $\mathcal{H}(\mathbf{\Gamma})$
- ↪ compute the equations of motion $\dot{\mathbf{\Gamma}}$
- ↪ find the conserved quantities
- ↪ define the distribution function $f(\mathbf{\Gamma})$





Partition function:

$$Z = \zeta \int dv_{\Gamma} f(\Gamma) = \zeta \int d\Gamma e^{-w(\Gamma)} \prod_{k=1}^{n_c} \delta(\Lambda_k(\Gamma) - C_k)$$

Generalized Liouville theorem $\rightarrow$

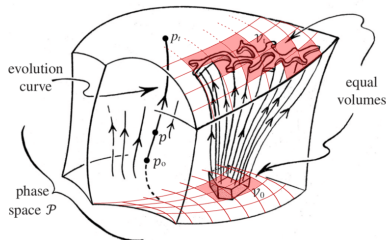
$\rightarrow$ **invariant volume form in the phase space**

$$dv_{\Gamma} = e^{-w(\Gamma)} d\Gamma$$

Exponential term $\rightarrow$

$\rightarrow$ phase space geometry information

$$w(\Gamma) = \int \kappa[\Gamma(t)] dt = \int \frac{\partial}{\partial \Gamma} \cdot \dot{\Gamma}(t) dt$$



Phase space compressibility:

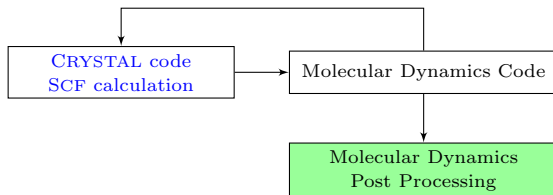
$$\kappa[\Gamma(t)] = \nabla_{\Gamma} \cdot \dot{\Gamma}(t)$$

General procedure to define the form of the partition function:

- $\rightarrow$ on the base of the conserved quantities $\rightarrow$ find the distribution function $f(\Gamma)$
- $\rightarrow$ on the base of the equations of motion $\rightarrow$ find the phase space compressibility $\kappa[\Gamma(t)]$
- $\rightarrow$ find the square root of the determinant of the metric of the phase space $\exp[-w(\Gamma)]$
- $\rightarrow$ use $f(\Gamma)$ and dv_{Γ} to define the partition function Z

Ab initio Molecular Dynamics : Post Processing

$$\begin{cases} \hat{\mathcal{H}}_e(\boldsymbol{\xi}; \mathbf{q}) \Psi_{e,0}(\boldsymbol{\xi}; \mathbf{q}) = E_{e,0}(\mathbf{q}) \Psi_{e,0}(\boldsymbol{\xi}; \mathbf{q}) & \text{where } E_{e,0} = \min_{\Psi_{e,0}} [\langle \Psi_{e,0} | \hat{\mathcal{H}}_e | \Psi_{e,0} \rangle] \\ \mathbf{f}_\mu(\Gamma(t)) + m_\mu \ddot{\mathbf{q}}_\mu(t) = \mathbf{F}_\mu = -\nabla_{\mathbf{q}_\mu} E_{e,0}(\mathbf{q}) & \mu = 1, \dots, N \end{cases}$$



The Post Processing properties currently implemented in the CRYSTAL code are

- ↗ the **Pair Correlation Function**
- ↗ Calculation of Velocity Autocorrelation Function useful to compute
 - ↗ the **Diffusion Coefficient**
 - ↗ the **Power Spectrum (Vibrational Density of States)**

Post Processing of AIMD Trajectory - Pair Correlation Function (PCF)

The **local structural environment** can be studied by means of the PCF given by

$$\text{Partial PCF (atomic species } \alpha \neq \sigma \text{): } g_{\alpha\sigma}(\mathbf{q}) = \frac{N}{\rho N_{\alpha} N_{\sigma}} < \sum_{\mu=1}^{N_{\alpha}} \sum_{\nu=1}^{N_{\sigma}} \delta(\mathbf{q} + \mathbf{q}_{\nu} - \mathbf{q}_{\mu}) >$$

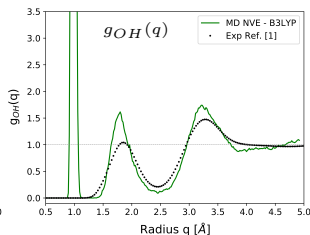
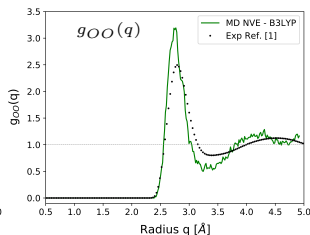
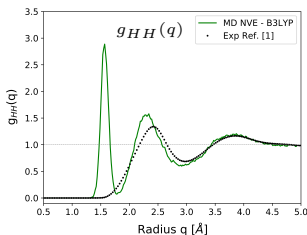
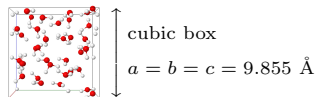
$$\text{Asymptotic behavior: } \lim_{\|\mathbf{q}\| \rightarrow \infty} g_{\alpha\sigma}(\mathbf{q}) = 1 - N^{-1}$$

32 H₂O molecules (96 atoms) 3D system and density $\rho = 1.000 \text{ g/cm}^3$

Functional : B3LYP

MD NVE with $\bar{T} = (397.80 \pm 24.13) \text{ K}$

$\tau P = 1.00 \text{ ps}$ ($\tau = 0.25 \text{ fs}$)



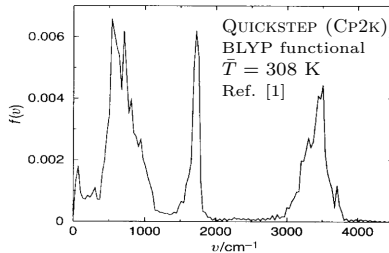
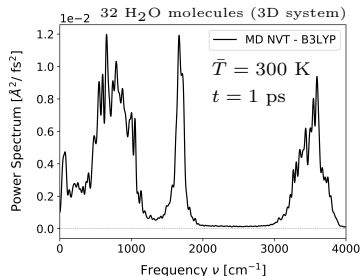
Post Processing of AIMD Trajectory - Power Spectrum

The **power spectrum** of an atomic system can be obtained from the **nuclear velocities** $\mathbf{v} = \{v_\alpha = p_\alpha/m_\alpha\}$ ($\alpha = 1, \dots, Nd$) collected during a molecular dynamics simulation

$$P(\omega) = \mathcal{F} \left[\lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} C_v(\tau) d\tau \right] \\ = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{\alpha=1}^{Nd} \left| \int_{-T/2}^{T/2} v_\alpha(t) e^{-i\omega t} dt \right|^2$$

with $C_v(\tau) \equiv \langle \mathbf{v}(t) \cdot \mathbf{v}(t + \tau) \rangle$

- ↪ The power spectrum (the vibrational density of states) can be easily computed using the Wiener-Khinchin theorem
- ↪ It describes the **collective motion of all the atoms** in the system during a molecular dynamics simulation



Ab initio Molecular Dynamics in the CRYSTAL code

- ↗ The **ab initio molecular dynamics** implemented in the CRYSTAL code demonstrates capable of performing reliable and efficient calculations in conjunction with **hybrid functionals**
- ↗ The **post processing analysis** of molecular dynamics trajectories permits to compute the pair correlation function, the diffusion coefficient and the power spectrum of the system
- ↗ Extensions to the molecular dynamics module include the **fast inertial relaxation engine approach** as structural optimization method for molecular and periodic systems

References

C. Ribaldone and S. Casassa, AIP Advances 12, 015323 (2022)

C. Ribaldone and S. Casassa, J. Chem. Theory Comput. 20, 3954-3975 (2024)

